

AE 429: Space Environment
Final Project
Due: April 30, 2026, 12:15pm
Total: 100 points

This is a partner project (groups of two only). Your job is to write a mission environmental analysis and test plan for your assigned mission (separate document). The plan must be justified using numerical analysis and concepts learned throughout the semester (perturbations, thermal, radiation, plasma, neutral environment, vibrations, debris, etc.).

At a minimum, provide the following sections and analysis:

1. Introduction

Purpose, mission overview, orbit, duration, constraints, and any assumptions or standards relevant to your spacecraft/mission.

2. System Description

Satellite architecture, major subsystems, interfaces, and key components. Include a 5x5 stoplight risk matrix summarizing major mission risks and how your test plan reduces risks.

3. Space Environment Analysis

For a minimum of four major environments (chosen from those covered in class):

- Characterize the environment of your orbit (be specific).
- Provide numerical estimates using equations learned in class where appropriate. You must clearly show the equation used, define variables, and state assumptions. Random numbers or values taken without justification will not receive credit.
- After computing a numerical value, you must explain what the result means physically (e.g., is it large/small, dangerous/negligible, what subsystem is affected). Simply reporting a number is not sufficient.
- Identify potential failure modes. When possible, relate these directly to your numerical results (i.e., what level or threshold leads to failure).
- Give mitigation strategies and justify them. You must explain how the mitigation reduces risk (qualitatively or quantitatively). Generic statements without explanation will not receive full credit.
- Where you can, include a simple model or simulation (e.g., thermal balance, vibration response, charging estimate, etc.). Clearly state assumptions, inputs, and interpret the results. Use of tools such as SPENVIS, STK, MATLAB, or similar is encouraged.

4. Test Objectives and Environmental Test Plan

State the purpose of environmental testing and explain how the planned tests verify performance, reduce risk, and address the hazards identified in your environment analysis.

Provide a summary of each test, including what environment or risk it addresses and the expected outcome.

Note: The goal of this project is to demonstrate understanding of the physical environments and their impact on spacecraft design. Reports that only list equations and numbers without interpretation or justification will receive limited credit.